25 In a circuit, two identical lamps are in parallel. The parallel pair of lamps is in series with a resistor and a battery.

The battery does 120 J of work moving 50 C of charge around the circuit in time *t*.

In each lamp, 40 J of work is done by the charge that passes through it in time *t*.

The electromotive force (e.m.f.) of the battery is *E* and the potential difference (p.d.) across one lamp is *V*.

Which statement about *E* and *V* is correct?

A *E* is 2.4 V and *V* is 0.80 V.

B *E* is 2.4 V and *V* is 1.6 V.

C *E* is 4.8 V and *V* is 2.4 V.

D *E* is 4.8 V and *V* is 4.8 V.